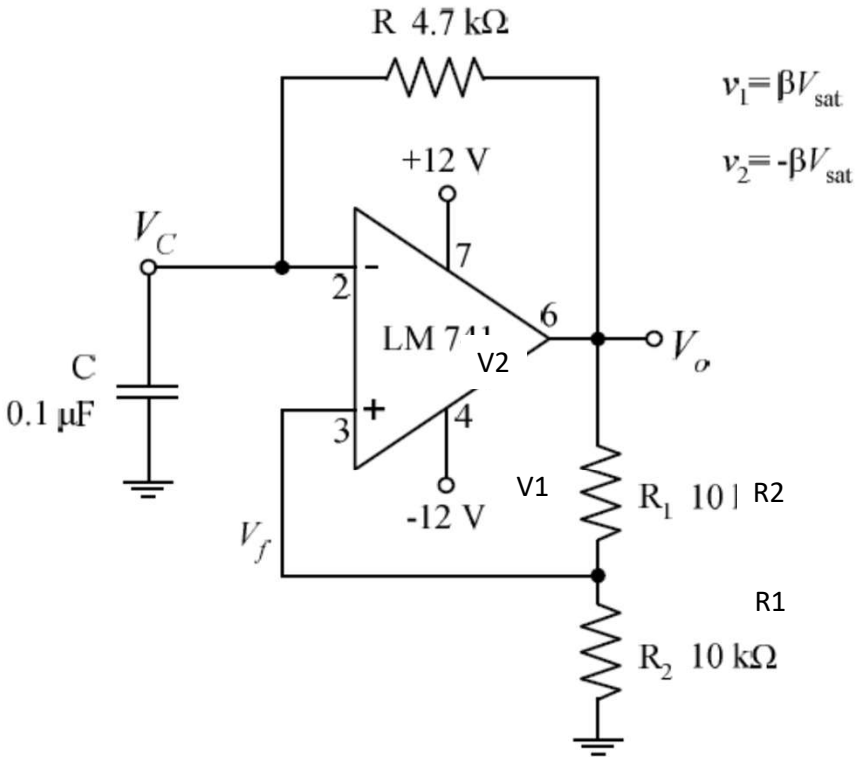
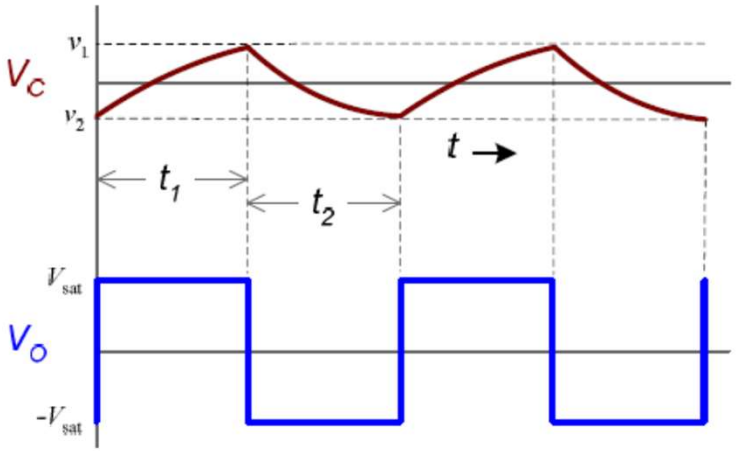
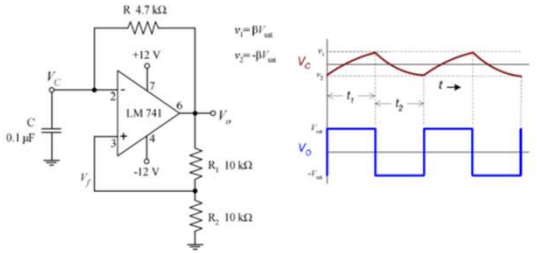
Waveform Generators

Square wave Oscillator/Generator





Square wave Oscillator/Generator



- The basic square wave oscillator is based on the charging and discharging of a capacitor.
- Op-amps inverting input is the capacitor voltage and the noninverting input is a portion of the output fed back through resistors R1and R2(refer figure 1).
- When the circuit is first turned on, the capacitor is uncharged, and thus the inverting input is at 0V. However voltage at noninverting terminal is small positive voltage (function of output offset voltage)
- This makes the output a positive maximum, and the capacitor begins to charge towards voltage at V_O through resistor R.

Square wave Oscillator/Generator

- When the capacitor voltage reaches a value equal to the feedback voltage (V_f) on the non-inverting input, the op-amp switches to the maximum negative state. Vout = -Vsat

Voltage across R1 is also negative

- At this point, the capacitor begins to discharge from $+V_f$ towards $-V_f$.
- When the capacitor voltage reaches $-V_f$, the op-amp switches back to the maximum positive state.
- This action repeats and a square wave output voltage is obtained.

Square wave Oscillator/Generator

V1 = $\frac{R1}{R1+R}(-Vsat)$

V1 = $\frac{R1}{R1+R}(+Vsat)$

Expression for period is

$$T = 2RC\ln\frac{1+\beta}{1-\beta} \qquad \text{where } \beta = \frac{R_2}{R_1+R_2}$$

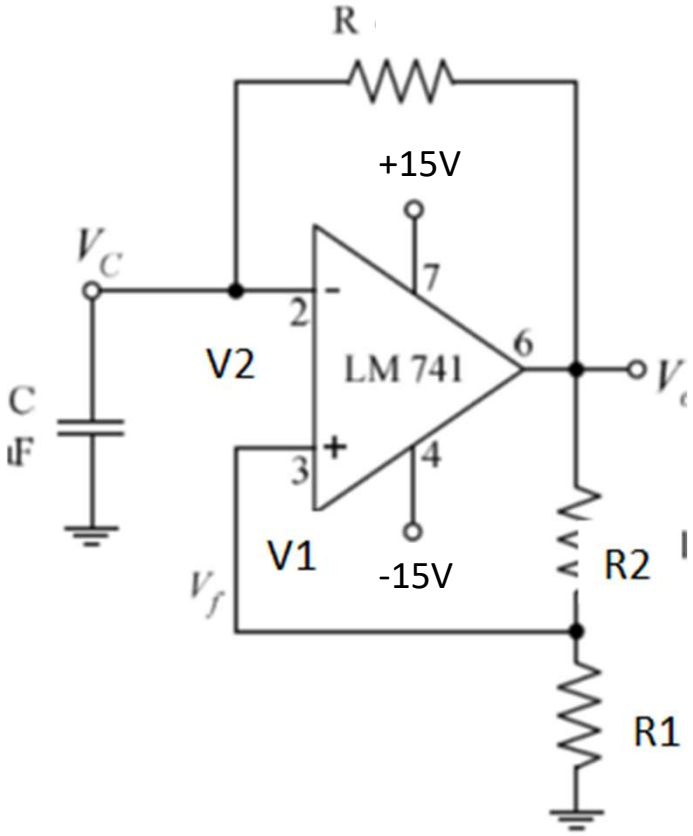
If $R_1 = R_2$, the equation for period reduces to $T = 2RC\ln 3$

The frequency of oscillation, $f = \frac{1}{2RC\ln 3}$

If R2 = 1.16R1, then F = 1/ 2RC

- Design square wave Oscillator(Generator), so that Fo = 1 kHz. The op-amp is 741 with dc supply voltage = ±15 V.

- Use R2 = 1.16 R1, Fo = 1/ (2RC)
- Let R1 = 10 kΩ, so R2 = 11.6 kΩ
- Assume/ let/Choose C = 0.05µF
- Fo = 1/ (2RC)
- 1 kHz = 1 / (2 R 0.05µF)
- R = 10 kΩ



Triangular wave Generator

- This circuit (figure 2) uses two operational amplifiers. Op-amp A1 functions as a comparator and the op-amp A2 as an integrator. Comparator compares the voltage at point P continuously with respect to the voltage at the inverting input; which as at ground potential.
- When the voltage at P goes slightly below zero, the output of A1 will switch to negative saturation. Suppose the output of A1 is at positive saturation $+V$sat.
- Since this voltage is the input of the integrator, the output of A2 will be a negative going ramp.
- Thus, one end of the voltage divider R1-R2 is at $+V$sat and the other at the negative going ramp.

Triangular wave Generator

- At time $t = t1$, when the negative going ramp attains value of $-V$ramp the effective voltage at point P becomes slightly less than 0 V. This switches output of A1 from positive saturation to negative saturation level $-V$sat.
- During the time when the output of A1 is at $-V$sat, the output of A2 increases in positive direction.
- At the instant $t = t2$, the voltage at point P becomes just above 0 V, thereby switching the output of A1 from $-V$sat to $+V$sat.
- The cycle repeats and generates a triangular waveform.

Triangular wave Generator

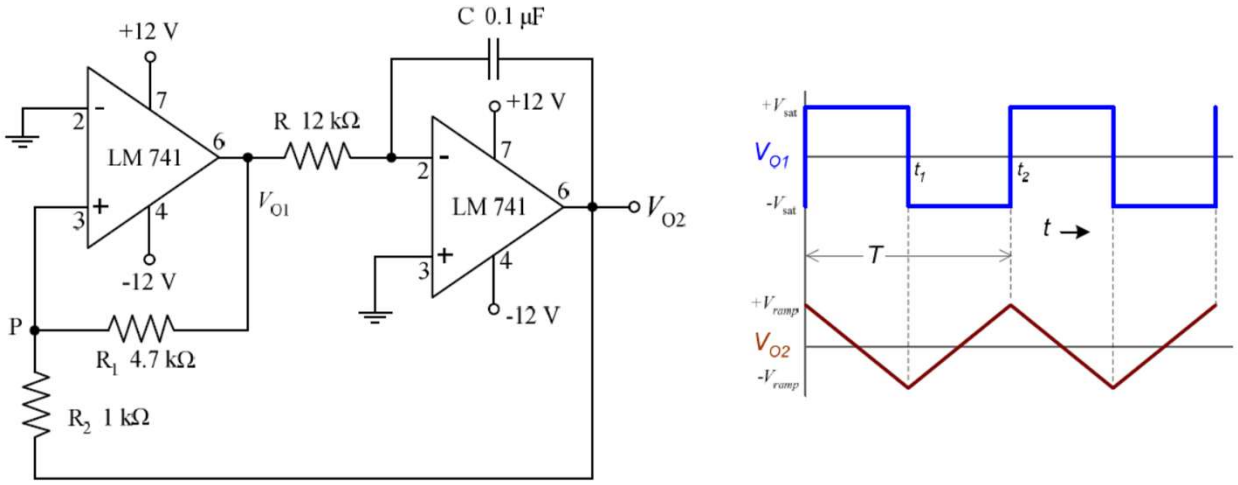


Fig 2. Triangular wave generator and waveforms

Triangular wave Generator

At $t = t_1$ $\quad \dfrac{-V_{ramp}}{R_2} = -\dfrac{+V_{sat}}{R_1}$ $\quad$ ie. $\quad -V_{ramp} = -\dfrac{R_2}{R_1}(+V_{sat})$

Similarly, at $t = t_2$ $\quad +V_{ramp} = -\dfrac{R_2}{R_1}(-V_{sat})$

The peak to peak output of the triangular wave is

$$V_{O(pp)} = +V_{ramp} - (-V_{ramp}) = 2\frac{R_2}{R_1}V_{sat}$$

During the period 0-t_1, The integrator functions as below.

$$V_{O(pp)} = \frac{1}{RC}\int_0^{T/2}(-V_{sat})\,dt = \left(\frac{V_{sat}}{RC}\right)\left(\frac{T}{2}\right)$$

Then, $\quad T = 2RC\left(\dfrac{V_{O(pp)}}{V_{sat}}\right)$

Substituting for $V_{O(pp)}$ $\quad T = \dfrac{4RCR_2}{R_1}$

Then, frequency of oscillation, $f = \dfrac{R_1}{4RCR_2}$

Sawtooth Wave Generator

- The difference between the triangular and sawtooth waveform is that the rise time of the triangular wave is always equal to its fall time while in sawtooth wave generator, rise time may be much higher than its fall time or vice versa.
- The triangular wave generator can be converted to a sawtooth wave generator by injecting a variable dc voltage into the noninverting terminal of the integrator.
- This can be done by using a potentiometer as shown in figure 3. When the wiper of the potentiometer is at the centre, the output will be a triangular wave since the duty cycle is 50%.
- If the wiper moves towards $-V$, the rise time of the sawtooth becomes longer than the fall time. If the wiper moves towards $+V$, the fall time becomes more than the rise time.

Sawtooth Wave Generator

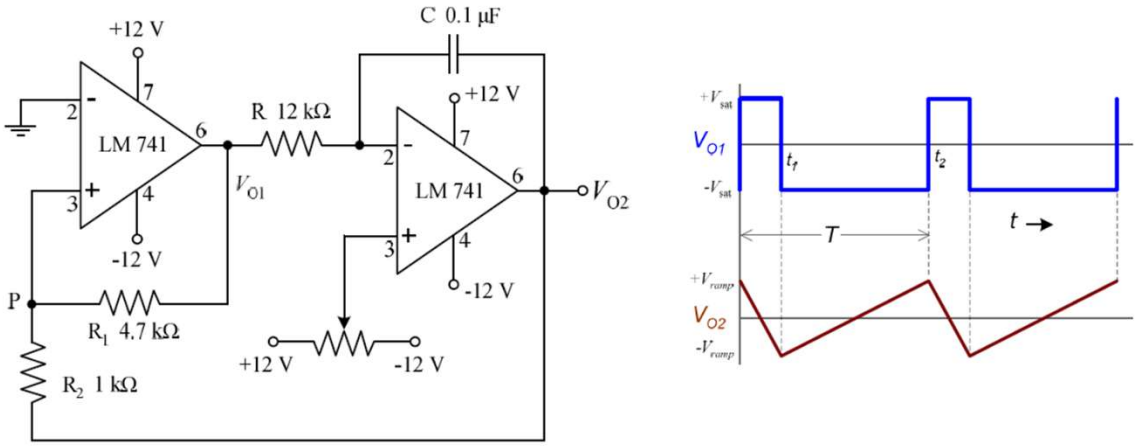


Fig 3. Sawtooth wave generator and waveforms